

Citizen Identity Management ACID Documentation

Transaction Scenario 1: ID Card Application Process

When a citizen applies for a new ID card, the system does several steps.

First, it saves the application.

Then it records the payment.

After payment is confirmed, it updates the application status.

If approved, the system issues a new ID card and deactivates the old active card.

Atomicity

All these steps work as one single process.

If anything fails (like payment failure), everything is cancelled.

No half-done data is saved.

Consistency

The system only saves valid data.

A citizen must exist before applying.

Dates must make sense (expiry date after issue date).

Only one active ID card is allowed per citizen.

These rules keep the database correct.

Isolation

If many officers are working at the same time, their work does not disturb each other.

Each process runs separately and safely.

Durability

Once everything is completed and saved, the data stays safe.

Even if the system crashes or power goes out, the data is not lost.

Transaction Scenario 2: Marriage Registration

When two citizens register their marriage, the system saves their marriage details.

Sometimes it also updates their address.

Atomicity

Marriage registration and any related updates are treated as one process.

If one part fails, nothing is saved.

Consistency

The same person cannot be both husband and wife.

Duplicate marriages are not allowed.

All data must follow logical rules.

Isolation

If different officers update records at the same time, they do not interfere with each other.

The system manages multiple users safely.

Durability

Once the marriage record is saved, it stays permanently in the system.

Concurrency Strategy

Many officers can use the system at the same time.

The system allows multiple users to work together without blocking everyone.

It makes sure one officer's changes do not conflict with another's work.

Users cannot see unfinished or temporary data from others.

This keeps the data accurate and the system running smoothly.